Causality Assessment

Task ID: task3 | Submitted by: Palak Shrivas

Objective

To evaluate the relationship between a drug and an adverse reaction using standard causality assessment methods (Naranjo scale and WHO-UMC system) and classify the reaction as Certain, Probable or Possible.

Case Study

On 01-Jun-2021, Raj was started on Amoxicillin 500 mg thrice a day for throat infection. On 02-Jun he developed itchy rashes over his body. On 03-Jun the dose was reduced to 250 mg thrice a day. On 04-Jun the rash disappeared. On 05-Jun the dose was increased back to 500 mg thrice a day. On 06-Jun he again developed generalized itchy rashes.

Method 1: Naranjo Adverse Drug Reaction Probability Scale

#	Question	Answer	Score
1	Previous conclusive reports on this reaction?	Yes – rash is a known amoxicillin reaction	+1
2	Event appeared after the suspected drug was given?	Yes – next day (02-Jun)	+2
3	Improved when drug was discontinued / dose reduced?	Yes – rash gone after dose reduced	+1
4	Reappeared when drug was re-administered?	Drug never stopped – no true rechallenge	0
5	Alternative causes?	Not mentioned – Do not know	0
6	Reaction with placebo?	Not done	0
7	Toxic drug level in blood?	Not measured	0
8	More severe with higher dose / less with lower dose?	Yes – 500 mg rash, 250 mg gone, 500 mg rash again	+1
9	Similar reaction to same/similar drug before?	Not known	0
10	Confirmed by objective evidence?	Not mentioned	0
	TOTAL SCORE		**5**

Score range: Definite ≥ 9, Probable 5–8, Possible 1–4, Doubtful ≤ 0. Total 5 = **PROBABLE**.

Method 2: WHO-UMC Causality Assessment

Rash occurred within a reasonable time after starting the drug, improved when the dose was reduced, and reappeared when the dose was increased. No other cause is given. A formal rechallenge was not done. This matches **Probable / Likely**.

Conclusion

Both Naranjo (score 5) and WHO-UMC classify the rash as **PROBABLE** to Amoxicillin.

Assumption: other causes and past reactions are not mentioned in the case, so they were scored 0.